

Apache NiFi is an open-source software project designed to automate the flow of data between software systems. It provides a web-based user interface for creating, monitoring, and controlling data flows. Developed by the NSA and now part of the Apache Software Foundation, NiFi is a powerful tool for data ingestion, routing, and processing.

Core Concepts

At its core, NiFi is based on a few key concepts:

- **FlowFile:** Represents a single piece of data moving through the system. A FlowFile consists of two parts: its content (the data itself) and its attributes (metadata about the data).
 - **Processor:** The basic building block for creating data flows. Each processor performs a specific task, such as reading data from a source, transforming it, or writing it to a destination.
 - **Connection:** Links processors together, creating a path for FlowFiles to travel. Connections act as queues, allowing data to be buffered and processed at a sustainable rate.
 - **Process Group:** A set of processors and their connections, which can be organized to create more complex data flows. Process Groups can be nested within each other, allowing for modular and reusable components.
 - **Data Provenance:** NiFi automatically records a detailed history of every FlowFile as it moves through the system. This provides a complete audit trail of the data's journey, which is invaluable for troubleshooting, compliance, and data governance.
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Key Differentiators

NiFi stands out from other data flow management tools in several ways:

- **User-Friendly Interface:** NiFi provides a drag-and-drop, web-based UI that makes it easy to create and manage data flows without writing any code. This visual approach makes it accessible to a wide range of users, from data engineers to business analysts.
- **Data Provenance:** The built-in data provenance feature is a major differentiator. It provides a complete, real-time record of every action performed on a piece of data, from its origin to its final destination. This level of detail is crucial for ensuring data quality, security, and compliance.
- **Back Pressure and Throttling:** NiFi is designed to handle high-volume data flows without being overwhelmed. It can automatically apply back pressure to slow down data sources when a downstream system is unable to keep up. This prevents data loss and ensures the stability of the entire system.
- **Security:** NiFi provides a secure platform for data exchange, with support for SSL/TLS encryption, user authentication, and role-based access control. It also supports secure communication with a wide range of external systems.

- **Extensibility:** While NiFi comes with a large number of built-in processors, it is also highly extensible. Custom processors and services can be developed in Java to meet specific needs, allowing for integration with virtually any system.
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Common Processors

NiFi has over 300 processors available, each designed to perform a specific task. Here are some of the most commonly used ones:

- **GetFile:** Reads files from a local or network-attached file system.
- **PutFile:** Writes files to a local or network-attached file system.
- **GetHTTP:** Fetches data from an HTTP or HTTPS URL.
- **PostHTTP:** Sends data to an HTTP or HTTPS URL.
- **ListenHTTP:** Starts an HTTP server and listens for incoming requests.
- **InvokeHTTP:** A more generic processor for making HTTP requests.
- **EvaluateJsonPath:** Extracts values from a JSON document using a JSONPath expression.
- **ReplaceText:** Replaces text in the content of a FlowFile using regular expressions.
- **RouteOnAttribute:** Routes a FlowFile to different connections based on the value of its attributes.
- **RouteOnContent:** Routes a FlowFile based on its content.
- **SplitJson:** Splits a JSON file into multiple FlowFiles, each containing a subset of the original data.
- **MergeContent:** Merges multiple FlowFiles into a single FlowFile.
- **ConvertRecord:** Converts data from one format to another, such as from CSV to Avro or JSON.
- **QueryDatabaseTable:** Executes a SQL query against a database and creates a FlowFile from the results.
- **PutSQL:** Executes a SQL statement to insert, update, or delete data in a database.
- **PublishKafka:** Sends messages to a Kafka topic.
- **ConsumeKafka:** Consumes messages from a Kafka topic.